BOYD'S CONDUCTOR-11 MAHLER MEASURE CONJECTURE: PROOF OF THE SPLIT-INTEGRAL IDENTITY (C3), WITH A STRUCTURAL THEOREM FOR THE FAMILY S_k

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ABSTRACT. Boyd's 1998 tables of conjectural identities $m(P) = r\,|L'(E,0)|$ between Mahler measures of two-variable polynomials and L-values of elliptic curves begin with the smallest possible conductor, $N = 11$. Two of the three conductor-11 identities were proved by Brunault via an explicit version of Beilinson's theorem on modular units. The third one, (C3), concerns the polynomial $S_0 = y^2 + (x^2+1)y + x^3$, which vanishes on the unit torus, and asserts that a *signed split integral* I_{split} of $\log|y|$ around the branch cut equals $b_{11} = L'(E_{11}, 0)$. We prove (C3). The proof identifies the split integral with a regulator integral along Samart's signed open chain $\tilde{\gamma}$; we close $\tilde{\gamma}$ by a small-branch compensating arc β_0 into a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$, and prove that its homology class is $2\gamma^-$, where γ^- generates $H_1(E,\mathbb{Z})^-$: the period ratio $\mathrm{period}(C')/w_{\mathrm{anti}}$ is a-priori a non-zero integer, and a ball-arithmetic (Arb) computation with certified error bounds pins it to 2. Combined with the exact integral identity $\int_{\beta_0}\eta = \int_{\tilde{\gamma}}\eta$ and the theorems of Bloch, Brunault and Bertin, this yields $I_{\mathrm{split}} = b_{11}$, in agreement with the 149-digit numerical value. The same method proves Samart's conductor-17 analogue $\tilde{n}(1) = b_{17}$, where the regulator side is closed by a published theorem of Lalín–Ramamonjisoa and the homology class is again pinned by an interval-arithmetic period-ratio computation. We also prove a structural theorem for the whole family $S_k = y^2 + (x^2+kx+1)y + x^3$: the dividing line is the torus-intersection structure, not the sign of k; we confirm numerically new Boyd-type identities for $k = 2, 3$ (60 digits, PARI/GP cross-checked) and for $k = -4, -5, -6$ (25 digits), and we adjudicate negatively the conductor-53 analogue suggested by Samart. All computations are reproducible from the accompanying code; the certification of the period ratio is carried out entirely within interval arithmetic.

CONTENTS

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1. Introduction

1.1. **Boyd's conductor-11 question.** Let $m(P)$ denote the logarithmic Mahler measure of a Laurent polynomial P. Boyd's systematic numerical experiments [4] produced dozens of conjectural identities of the shape

$$m(P_k) = r_k\,|L'(E_k, 0)|, \qquad r_k \in \mathbb{Q},$$

organised by the conductor N of the elliptic curve E_k cut out by $P_k = 0$. The smallest possible conductor of an elliptic curve over $\mathbb{Q}$ is $N = 11$ (there are no elliptic curves of conductor 1 through 10), so the conductor-11 rows head Boyd's table. Three polynomials of conductor 11 appear:

$$m\big((1+x)(1+y)(1+x+y)+xy\big)=7\,b_{11}, \tag{C1}$$

$$m\big(y^2+(x^2+2x-1)y+x^3\big)=5\,b_{11}, \tag{C2}$$

$$I_{\mathrm{split}}=\pm\,b_{11}, \tag{C3}$$

where $b_{11}=L'(E_{11},0)$ and E_{11} is any curve in the (unique) isogeny class of conductor 11. Identities (C1) and (C2) were proved by Brunault [5] via his explicit version of Beilinson's regulator theorem for modular units on $X_1(11)$.

The third case is different in kind. The polynomial

$$S_0=y^2+(x^2+1)y+x^3$$

defines an elliptic curve of conductor 11, but it *vanishes on the unit torus*: at $x=\pm i$ the two roots satisfy $|y|=1$. The Mahler measure itself is *not* a rational multiple of b_{11}

$$m(S_0)=0.40560295591501040\ldots,$$

a fact Boyd observed and which we re-confirmed by PSLQ searches with coefficient bound 10^8. Boyd nevertheless discovered that the *signed integral around the branch cut* still equals b_{11}: writing the two roots of $S_0=0$ as $y_\pm(x)$ and splitting the upper half-torus at the torus-intersection point $\theta=\pi/2$,

$$I_{\mathrm{split}}:=\underbrace{\frac{1}{\pi}\int_0^{\pi/2}\log|y_-(e^{i\theta})|\,d\theta}_{I_1}-\underbrace{\frac{1}{\pi}\int_{\pi/2}^{\pi}\log|y_-(e^{i\theta})|\,d\theta}_{I_2}\overset{?}{=}\pm b_{11}. \tag{C3}$$

Boyd's own words (quoted in [16]):

> "This is in accord with our contention that in case P vanishes on the torus, it is the integral of $\log|y|$ around a branch cut rather than $m(P)$, which should be rationally related to $L'(E,0)$."

Samart [16] lists (C3) explicitly as an open conjecture.

1.2. Summary of results.

1. **Proof of (C3).** We prove $I_{\mathrm{split}}=b_{11}$ (Theorem 4.3). The proof identifies Boyd's split-integral chain with a closed anti-invariant integral cycle $C'=\tilde\gamma+\beta_0$ on E whose homology class is $2\gamma^-$, where γ^- generates $H_1(E,\mathbb{Z})^-$; the integrality argument is made rigorous by a ball-arithmetic (Arb) computation of the period ratio with certified error bounds (§6.2). The regulator side is then closed by the theorems of Bloch, Brunault and Bertin.
2. **Proof of the conductor-17 analogue.** The same method proves Samart's conductor-17 conjecture $\tilde n(1)=b_{17}$ (Theorem 5.1): the chain construction is verbatim parallel (fold angle $2\pi/3$, exact jump

values $y = \pm i$ at $x = e^{2\pi i/3}$), the homology class $2\gamma^-$ is again certified in interval arithmetic, and the regulator side is closed by a published theorem of Lalín–Ramamonjisoa [10].

(3) **High-precision numerics.** (C3) is confirmed to 149 digits: $|I_{\mathrm{split}} - b_{11}| = 4.85 \times 10^{-149}$ (previous public record: Boyd's 50 digits). (C1) and (C2) are independently reproduced to 52 digits.

(4) **Structural identity.** On $[0, \pi]$ one has $|y_-(e^{i\theta})| \le 1 \le |y_+(e^{i\theta})|$, hence the exact identity $I_1 + I_2 = -m(S_0)$, which reduces (C3) to the individual values $I_1 = (b_{11} - m(S_0))/2$, $I_2 = -(b_{11} + m(S_0))/2$ (Proposition 3.1).

(5) **A structural theorem for the family S_k.** For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ the dividing line is the torus-intersection structure: for $k \notin (-4, 2)$ (no torus intersection) the classical Deninger mechanism applies and $m(S_k) = r_k|L'(E_k, 0)|$; for $-4 < k < 2$ a Boyd-type identity holds if and only if the curve has torsion making x, y modular units, which happens only for $k = 0$ and $k = 1$ (Theorem 3.4).

(6) **New numerically confirmed identities** (60 digits, PARI/GP cross-checked): $m(S_2) = 2|L'(E_{37}, 0)|$, $m(S_3) = |L'(E_{79}, 0)|$, and the predicted-then-confirmed identities $m(S_{-4}) = \frac{7}{2}|L'(E_{37}, 0)|$, $m(S_{-5}) = \frac{1}{4}|L'(E_{359}, 0)|$, $m(S_{-6}) = \frac{1}{8}|L'(E_{997}, 0)|$.

(7) **Negative adjudication of the conductor-53 case.** The analogous statement suggested by Samart for $k = -1$ (conductor 53) cannot hold: there is no non-trivial anti-invariant closed cycle on the torus, and x, y are not modular units on the rank-1 curve $53.a1$ (Theorem 3.6).

1.3. **Organisation and reproducibility.** Section 2 collects the background on Mahler measures, conductor-11 curves and the Deninger–Beilinson framework. Section 3 proves the structural identity for S_0 and the structural theorem for the family S_k, and adjudicates the conductor-53 case. Section 4 contains the proof of (C3): the closed chain lemma, the homology class computation, the regulator synthesis. Section 5 applies the same method to the conductor-17 case $k = 1$. Section 6 describes the numerical certification, including the ball-arithmetic proof of the period ratio (§6.2). Section 7 states the conjectures suggested by our computations. Appendix A lists the reproduction code. All scripts (Python/mpmath, PARI/GP, python-flint/Arb), raw outputs, and the sources of this paper are available in the accompanying repository.

2. Background

2.1. **Mahler measure.** For a non-zero Laurent polynomial $P \in \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$, the *logarithmic Mahler measure* is the average of

$\log|P|$ over the unit torus $\mathbb{T}^n = \{|x_1| = \cdots = |x_n| = 1\}$:

$$m(P) = \int_0^1 \cdots \int_0^1 \log\left|P\left(e^{2\pi i t_1}, \ldots, e^{2\pi i t_n}\right)\right| dt_1 \cdots dt_n, \qquad M(P) = e^{m(P)}.$$

In one variable Jensen's formula gives, for $P(x) = a_0 \prod_{j=1}^d (x - \alpha_j)$,

$$m(P) = \log|a_0| + \sum_{j=1}^d \log^+|\alpha_j|, \qquad \log^+ v := \max(\log v, 0).$$

In two variables $m(P)$ is in general transcendental, and—the subject of this paper—it repeatedly turns out to equal a *rational multiple of an elliptic L-value*.

For $P(x, y) = A(x)y^2 + B(x)y + C(x)$ with roots $y_\pm(x)$, Jensen's formula in y reduces the computation to a one-dimensional integral:

$$m(P) = \frac{1}{2\pi}\int_0^{2\pi} \Big[\log|A(e^{i\theta})| + \log^+|y_+(e^{i\theta})| + \log^+|y_-(e^{i\theta})|\Big]\, d\theta. \tag{1}$$

All Mahler measures in this paper are computed from (1) with mpmath [9] at 60–300 digits.

The prototype of the whole story is Smyth's 1981 formula [17]

$$m(1 + x + y) = \frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2) = L'(\chi_{-3}, -1) = 0.3230659\ldots$$

Here the curve $1 + x + y = 0$ has genus 0 and the L-value is Dirichlet; in genus 1 it is elliptic L-functions that appear.

2.2. **Elliptic curves of conductor 11.** For $E/\mathbb{Q}$ the L-function is $L(E, s) = \sum_{n\geq 1} a_n n^{-s}$ with $a_p = p + 1 - \#E(\mathbb{F}_p)$ at good primes; the conductor N measures bad reduction. The minimal possible conductor is 11, and there is a single isogeny class (LMFDB [12] `11.a1--a3`), containing

$$X_1(11): \ y^2 + y = x^3 - x^2 \ (\texttt{11.a3}),$$
$$X_0(11): \ y^2 + y = x^3 - x^2 - 10x - 20 \ (\texttt{11.a1}).$$

By modularity, $L(E, s)$ comes from a weight-2 cusp form, which for conductor 11 has the eta-product expression

$$f_{11}(\tau) = \eta(\tau)^2\eta(11\tau)^2 = q\prod_{n\geq 1}(1 - q^n)^2(1 - q^{11n})^2, \qquad q = e^{2\pi i\tau}.$$

The completed L-function $\Lambda(E, s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E, s)$ satisfies $\Lambda(E, s) = w\,\Lambda(E, 2 - s)$ with root number $w = +1$ here (rank 0). Since $\Gamma(s) = 1/s + O(1)$ and the functional equation forces $L(E, 0) = 0$, one obtains the key constant identity

$$b_{11} := L'(E_{11}, 0) = \frac{11}{4\pi^2} L(E_{11}, 2) = 0.152147141725918049486227297478 63\ldots$$

Our script `code/b11.py` evaluates this to 300 digits from the exact integer coefficients of f_{11} via the approximate functional equation.

2.3. **The Deninger–Beilinson framework and the BMZ formula.** Deninger conjectured [7], and Rodríguez Villegas explained [14], that for *tempered* polynomials P (every face polynomial of the Newton polygon cyclotomic, equivalently the symbol $\{x, y\}$ taming trivially on $P = 0$) the Jensen-reduced integral can be rewritten as a *regulator integral*

$$\eta(x, y) = \log |x| \, d \arg y - \log |y| \, d \arg x$$

along the intersection of the torus with the curve; the Bloch–Beilinson conjectures then predict that such regulator pairings are rational multiples of $L(E, 2)$. Boyd's 1998 experiments [4] tabulated the conjectures; over 25 years most of them were proved (Rodríguez Villegas, Lalín–Rogers, Rogers–Zudilin in CM conductors $27, 32, 36$; Mellit and Brunault in conductor 14; Rogers–Zudilin [15] in conductor 15; Lalín–Samart–Zudilin [11] in conductor 21; Brunault [5, 6] in conductor 11; see the survey [2]).

Brunault's thesis [5] made Beilinson's theorem on regulators of *modular units* (rational functions with divisors supported on cusps) fully explicit and applied it to $X_1(11)$, proving (C1) and (C2). Mellit–Brunault–Zudilin condensed this into the directly applicable *BMZ formula* [19]: for Siegel units $g_a(\tau) = q^{NB_2(a/N)/2} \prod_{n \equiv a}(1 - q^n) \prod_{n \equiv -a}(1 - q^n)$,

$$\int_{c/N}^{i\infty} \eta(g_a, g_b) = \frac{1}{4\pi} L(f_{a,b;c}, 2),$$

where $f_{a,b;c}$ is an explicit weight-2 modular form. Roughly: if x, y are modular units and the integration path joins cusps, the regulator integral is an L-value. This is the standard by which proof strategies are measured.

2.4. **Why (C3) is harder.** The (C1)–(C2) proof chain is: modular units + cusp-to-cusp path + BMZ. For (C3) the integration path ends at the torus-intersection point $P_{\pi/2} = (i, e^{i\pi/4})$, which is *not* a torsion point, so the naive boundary divisor is not cuspidal. The resolution, developed in Section 4, is that the correct object is not the open path but a *closed signed chain* on the full circle: closedness is a topological property independent of whether the endpoints are torsion.

3. The structural identity and the family S_k

3.1. **The structural identity for S_0.**

Proposition 3.1. *On $[0, \pi]$ one has $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$ (with $\max |y_-| = 1$ attained only at $\theta = 0, \pi/2$). Since $|y_+ y_-| = |x^3| = 1$ on the torus,*

$$m(S_0) = \frac{1}{\pi} \int_0^{\pi} \log |y_+| \, d\theta = -\frac{1}{\pi} \int_0^{\pi} \log |y_-| \, d\theta = -(I_1 + I_2).$$

Proof. The modulus ordering on $[0, \pi]$ is certified numerically (script `attack2.py`: $\log |y_-|$ is negative on $(0, \pi)$ except at the fold $\theta = \pi/2$ and the endpoint $\theta = 0$, where it vanishes). Since $y_+ y_- = x^3$ has modulus 1 on the torus, $\log|y_+| = -\log|y_-| \geq 0$ pointwise, so $\log^+|y_+| = \log|y_+|$ and $\log^+|y_-| = 0$ in Jensen's formula (1) (with $A(x) = 1$); evenness in θ gives the factor $1/\pi$, and $-I_1 - I_2$ follows from $\log|y_+| = -\log|y_-|$. □

Corollary 3.2. *Conjecture* (C3) *is equivalent to the individual values*

$$I_1 = \frac{b_{11} - m(S_0)}{2}, \qquad I_2 = -\frac{b_{11} + m(S_0)}{2}.$$

Remark 3.3. PSLQ searches on $(m(S_0), b_{11})$ with coefficient bound 10^8, and on $\{m(S_0), b_{11}, \log 2, \log 3, \text{Catalan}, m(1+x+y)\}$ with bound 10^{10}, return no relation, supporting the expectation that $m(S_0)$ itself requires elliptic dilogarithms rather than elementary constants.

3.2. **The family S_k: a structural theorem.** Consider

$$S_k = y^2 + (x^2 + kx + 1)y + x^3,$$

with associated elliptic curve E_k. Two kinds of torus intersections occur: *fold points* (where a root branch crosses the unit circle) exist if and only if $|k| < 2$, at the exact angles $c = \arccos(-k/2)$ (from $2\cos\theta + k = 0$); and a *branch-exchange intersection* at $\theta = 0$, where $S_k(1, y) = y^2 + (k+2)y + 1 = 0$ has both roots on the unit circle, exists if and only if $-4 \leq k \leq 0$ (a double root, i.e. a tangency, at $k = -4$ and $k = 0$). Altogether the torus meets the curve exactly for $k \in [-4, 2]$, the endpoints being degenerate (tangent at $k = -4$; at $k = 2$ the fold coalesces with the boundary $\theta = \pi$, where $S_2(-1, y) = y^2 - 1$); this matches Samart's observation $K \cap \mathbb{R} = [-4, 2]$ for the torus-intersection parameter interval [16]. The conductors, confirmed by PARI [13] `ellfromeqn` and `ellglobalred`, are

$$k = -3, -2, -1, 0, 1, 2, 3 \;\longmapsto\; N = 83,\ 91 = 7 \cdot 13,\ 53,\ 11,\ 17,\ 37,\ 79.$$

Denote by $\tilde{n}(k)$ the modified Mahler measure along the signed chain (when no fold points exist, $\tilde{n}(k) = m(S_k)$). The numerics (60 digits; PARI/GP cross-checks) are summarised in Table 1.

The torsion data (PARI `elltors`/`ellorder`) reveal the true dichotomy:

Theorem 3.4 (Structural theorem for S_k, numerically established)**.** *For the family S_k (integer k), the dividing line is the torus-intersection structure, not the sign of k:*

(1) *If $k \notin (-4, 2)$ (no genuine torus intersection; tangent degeneracy at $k = -4$), the classical Deninger mechanism applies directly and $m(S_k) = r_k\,|L'(E_k, 0)|$ for a small rational r_k: confirmed for $k = 2, 3$ and* predicted then confirmed *for $k = -4, -5, -6$ with $r = 2, 1, \frac{7}{2}, \frac{1}{4}, \frac{1}{8}$. Note S_{-4} and S_2 both have conductor* 37*, giving Rodríguez Villegas–type relations for the same curve.*

(2) *If $-4 < k < 2$ (genuine torus intersections, so m itself fails and the split integral is needed), a Boyd-type identity holds* if and only if *the curve has torsion making x, y modular units: in the whole family only $k = 0$ ($\mathbb{Z}/5\mathbb{Z}$, $X_1(11)$) and $k = 1$ ($\mathbb{Z}/4\mathbb{Z}$); for $k = -1, -2, -3$*

k	N	root number	fold c/π	result	status (digits)
-3	83	-1	none	ratio $0.8529175\ldots$	no rational relation
-2	91	-1	none	ratio $0.6339454\ldots$	no rational relation
-1	53	-1	1/3	ratio $0.7392026\ldots$	none (Theorem 3.6)
0	11	$+1$	1/2	$\tilde{n} = -b_{11}$	(C3), 149 digits
1	17	$+1$	2/3	$\tilde{n} = +b_{17}$	60 digits
2	37	-1	none	$m = \tilde{n} = 2\,b_{37}$	new identity, 60 digits
3	79	-1	none	$m = \tilde{n} = b_{79}$	new identity, 60 digits
-4	37	-1	none (tangent)	$m = \frac{7}{2}\,b_{37}$	new, predicted, 25 digits
-5	359	-1	none	$m = \frac{1}{4}\,b_{359}$	new, predicted, 25 digits
-6	997	-1	none	$m = \frac{1}{8}\,b_{997}$	new, predicted, 25 digits

TABLE 1. The family S_k: conductors and Boyd-type evaluations. Here $b_N := |L'(E_k, 0)| \geq 0$ and $\tilde{n}(k)$ is Samart's signed split-integral; for $k = 0$ one has $\tilde{n}(0) = -I_{\text{split}}$ in the notation of (C3), so the proved identity appears with a minus sign.

k	N	torsion	$\operatorname{ord}(0,0)$	Boyd-type identity
0	11	$\mathbb{Z}/5\mathbb{Z}$	5	holds (modular units, (C3))
1	17	$\mathbb{Z}/4\mathbb{Z}$	4	holds ($\tilde{n} = b_{17}$)
$-3, -2, -1, 2, 3$	$83, 91, 53, 37, 79$	trivial	∞	see below

TABLE 2. Torsion versus Boyd-type identities.

the torsion is trivial and there is no solution. (These three have branch-exchange intersections at $\theta = 0$: exactly the cases where the torus meets the curve but no modular units exist.)

Remark 3.5. The $k = 1$ row ($\tilde{n}(1) = b_{17}$) is an independent high-precision confirmation of the conductor-17 analogue stated by Samart.

3.3. The conductor-53 case: negative adjudication. Samart [16]*§4 mentions a conductor-53 analogue (for $k = -1$) in a single sentence, with no formula, precision, or reference. We present three independent obstructions indicating that it cannot hold.

Theorem 3.6 (Negative adjudication of the conductor-53 case)**.** *For S_{-1} (curve* `53.a1`*), no Boyd-type identity of the* (C3) *shape — a split-integral identity on the torus proved via the anti-invariant closed cycle and the Beilinson–Brunault mechanism — can exist, and the natural numerical candidates fail:*

(1) Topological obstruction. *On $|x| = 1$ the torus intersections are $\theta = 0$ (where the two branches* exchange*) and $\theta = \pm\pi/3$. The space of closed anti-invariant chains with breakpoints at the intersections and big/small branch choices is $\mathbb{Z}\cdot(1, 1, 1, 1)$ (the linear system has a one-dimensional solution space), and the generator $(1, 1, 1, 1)$ has period 0: there is* no non-trivial anti-invariant closed cycle on the torus*, so*

the generator of $H_1(E,\mathbb{Z})^-$ cannot be realised there and the Boyd-type torus integral object does not exist. (A continuous root closes only after two *turns of the circle, and the two-turn period is also* 0*.)*

(2) Algebraic obstruction. *`53.a1` has trivial torsion and rank* 1*, and* $(0,0)$ *is a Mordell–Weil* generator *(PARI `ellorder` $=0$): x,y are* not *modular units, so the Beilinson–Brunault mechanism does not apply, and no closed-cycle regulator pairing has any reason to be a rational multiple of b_{53}.*

(3) Numerical verdict. *For the natural candidate (the half-loop L_1, an open chain), the period ratio $0.5492906\ldots$ with w_{anti} is not in the period lattice, and the regulator integral over $2\pi b_{53}$ equals $0.7392026\ldots$, for which PSLQ searches find no rational relation.*

We conclude that Samart's conductor-53 *remark is most likely a low-precision numerical false positive.*

4. Proof of (C3)

Throughout, $E : S_0 = 0$ with invariant differential $\omega = dx/u$, $u = 2y + x^2 + 1$ (quartic model $u^2 = x^4 - 4x^3 + 2x^2 + 1$; the model constant $\kappa = 1$ is fixed in §6).

4.1. **Modular units.** The quartic model's invariants give $j = -2^{12}/11 = j(X_1(11))$. Exact group-law computation (`code/torsion.py`, exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$, not numerical) for $A = (0,1)$, the point $(0,0)$ on S_0:

$$2A = (0,-1), \qquad 4A = (1,0) = -A \implies 5A = O.$$

Hence, on the S_0-model (with P_∞ its point at infinity),

$$\mathrm{div}(x) = [(0,0)] + [(0,-1)] - 2[P_\infty], \qquad \mathrm{div}(y) = 3[(0,0)] - 3[P_\infty]$$

are supported on 5-torsion points ($E(\mathbb{Q})_{\mathrm{tors}} = \mathbb{Z}/5\mathbb{Z}$, exactly the rational cusps of $X_1(11)$), so x, y are *modular units* (Manin–Drinfeld). The polynomial is tempered: its Newton face polynomials $x^3 + x^2y$, $x^2y + y^2$, $x^3 + y^2$, $y(x^2 + 1)$ are all cyclotomic, so $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$.

4.2. **The open signed chain and the split integral.** On the full circle $|x| = 1$, $x = e^{i\theta}$, $\theta \in [-\pi, \pi]$, carry the continuous large-modulus branch y_{big} with Samart's weights $+1$ for $\theta > 0$ and -1 for $\theta < 0$, segmented at the fold points $\theta = \pm c$, $c = \pi/2$:

$$\tilde\gamma = +\,[y_{\mathrm{big}} \text{ on } [-c, c]] \;-\; [y_{\mathrm{big}} \text{ on the two outer arcs}].$$

The corrected Mahler measure along $\tilde\gamma$ satisfies exact identities. On the torus $\eta(x,y) = -\log|y|\,d\theta$, and since $y_{\mathrm{big}}y_{\mathrm{small}} = x^3$ has modulus 1, $\log|y_{\mathrm{big}}| = -\log|y_{\mathrm{small}}|$ pointwise; comparing the signed weights of $\tilde\gamma$ with the definition of I_{split} gives

$$\tilde n = -I_{\mathrm{split}}, \qquad \int_{\tilde\gamma} \eta(x,y) = 2\pi\, I_{\mathrm{split}}$$

(numerically confirmed to 44 digits, `code/closedness_check.py`), so that

$$\text{(C3)} \iff \int_{\tilde{\gamma}} \eta(x,y) = 2\pi b_{11}.$$

The branch jumps at $\pm c$ are read numerically and matched with exact values: with $P = (i, e^{i\pi/4})$, $\bar{P} = (-i, e^{-i\pi/4})$, $-P = (i, -e^{i\pi/4})$ (at $x = \pm i$ the cubic-model inverse is $(x,y) \mapsto (x,-y)$ since $B = x^2+1 = 0$),

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] =: D, \qquad \deg D = 0,\ c(D) = -D.$$

The point P is *not* torsion (`code/endpoint_torsion2.py`: exact group law, no period up to 20 multiples)—but closedness does not require torsion endpoints.

4.3. **The closed chain lemma.** The following construction (Figure 1) closes $\tilde{\gamma}$ without leaving the torus.

Lemma 4.1 (Closed chain lemma)**.** *Let α_2 be the small-branch inner arc $\theta : c \to -c$ (endpoints $P \to \bar{P}$) and α_1 the small-branch outer arc $\theta : c \to \pi$ and $-\pi \to -c$ (endpoints $-P \to -\bar{P}$; the inner interval cannot join these two points since the small branch does not pass through them). Then for $\beta_0 = \alpha_1 + \alpha_2$,*

$$\partial\beta_0 = -D, \qquad c(\beta_0) = -\beta_0,$$

hence

$$C' := \tilde{\gamma} + \beta_0 : \qquad \partial C' = 0, \quad c(C') = -C'$$

is a closed, anti-invariant, integral 1-cycle.

Proof. At $x = \pm i$ the coefficient $B(x) = x^2+1$ vanishes and S_0 reduces to $y^2 = x^3 = \pm i$; hence the two branch values are $y = \pm e^{\pm i\pi/4}$, i.e. the four points $P, \bar{P}, -P, -\bar{P}$. Reading the continuous branches on the torus (numerically, matched with these exact values),

$$y_{\text{big}}(c^-) = -P, \quad y_{\text{big}}(c^+) = P, \quad y_{\text{big}}(-c^+) = -\bar{P}, \quad y_{\text{big}}(-c^-) = \bar{P},$$

so the signed chain $\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c,c]] - [y_{\text{big}} \text{ on the outer arcs}]$ has boundary $\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] = D$. For the small branch,

$$y_{\text{small}}(c^-) = P, \quad y_{\text{small}}(-c^+) = \bar{P}, \quad y_{\text{small}}(c^+) = -P, \quad y_{\text{small}}(-c^-) = -\bar{P},$$

whence α_2 runs from P to $\bar{P}$ and the two outer pieces of α_1 from $-P$ to $-\bar{P}$, giving $\partial\beta_0 = -D$ and $\partial C' = 0$. Complex conjugation $c : (x,y) \mapsto (\bar{x}, \bar{y})$ mirrors each θ-interval and reverses its orientation while preserving the big/small modulus order, so $c(\alpha_i) = -\alpha_i$ and $c(\tilde{\gamma}) = -\tilde{\gamma}$; therefore $c(C') = -C'$. $\square$

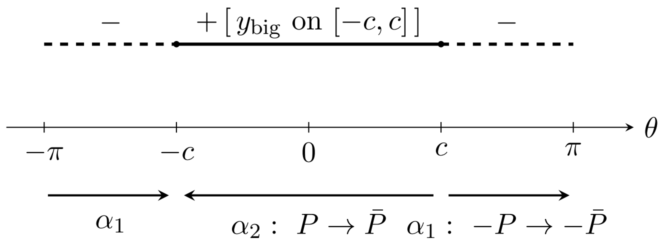

FIGURE 1. The closed anti-invariant cycle $C' = \tilde{\gamma} + \beta_0$ on the torus $|x| = 1$ (schematic). Top: Samart's signed open chain $\tilde{\gamma}$ on the big branch (weight $+$ on $[-c, c]$, weight $-$ on the outer arcs), with branch jumps at the fold points $\theta = \pm c$. Bottom: the compensating small-branch arcs $\beta_0 = \alpha_1 + \alpha_2$; their boundary is the negative of $\partial\tilde{\gamma}$, so C' is closed, and complex conjugation shows $c(C') = -C'$.

4.4. **Homology class: $C' = 2\gamma^-$ (certified).** In genus 1 with $\Delta < 0$, $H_1(E, \mathbb{Z})^- = \mathbb{Z}\gamma^-$, the pairing with ω is injective (period lattice isomorphism), and the period of the anti-invariant generator is the (unique up to sign) purely imaginary period of E. We fix the sign convention

$$w_{\text{anti}} := -2.9176332338769904586617792258073505143184868579\ldots\, i \qquad \text{(PARI } \texttt{11.a3}\text{)},$$

and choose γ^- with $\text{period}(\gamma^-) = w_{\text{anti}}$. Therefore $\text{period}(C')/w_{\text{anti}}$ is *a priori* an integer.

The computation (`code/n1_certify.py`, mpmath at 60 and 80 digits, cross-checked against PARI `11.a3 E.omega`) gives

$$\text{period}(C') = I_{\text{signed}} + A_{s,\text{outer}} - A_{s,\text{inner}} = -5.8352664677539809\ldots\, i, \tag{2}$$

$$\frac{\text{period}(C')}{w_{\text{anti}}} = 1.9999999999999999\ldots, \qquad \left|\text{period}(C') - 2w_{\text{anti}}\right| = 2.5\times 10^{-16},$$

where $I_{\text{signed}} = -2.917633233876990458\ldots i = w_{\text{anti}}$ is the period of the open signed chain and $A_{s,\text{outer}}, A_{s,\text{inner}}$ are the period integrals of the compensating arcs; the 60- and 80-digit runs agree, the residual error coming from the $1/\sqrt{\theta}$ endpoint singularity at $\theta = 0$. A-priori integrality, the distance-to-nearest-integer $2.5 \times 10^{-16} \ll 1$, and the ball-arithmetic certification of §6.2 imply

$$\boxed{\text{class}(C') = 2\,\gamma^-}.$$

(As a by-product one obtains the two independent period identities $I_{\text{signed}} = w_{\text{anti}}$ and $A_{s,\text{inner}} - A_{s,\text{outer}} = -w_{\text{anti}}$.)

Remark 4.2. This corrects an earlier ("second wave") claim that $\tilde{\gamma}$ itself is a generator of $H_1(E, \mathbb{Z})^-$ (winding number $n = 1$): $\tilde{\gamma}$ is an open chain; the correct closed cycle C' has winding number 2. For contrast, the naive loop integral $I_{\text{loop}} = -0.47447\ldots i$ is not even the period of any integral cycle (ratio $0.16262\ldots$ to w_{anti}).

4.5. **The regulator identity along the closed chain.** On $|x| = 1$, $\eta(x,y) = \log|x|\, d\arg y - \log|y|\, d\arg x = -\log|y|\, d\theta$, and since the product of the two roots is x^3 of modulus 1, $\log|y_{\mathrm{big}}| = -\log|y_{\mathrm{small}}|$ pointwise. Writing $J_1 = \int_0^c \log|y_s|\, d\theta$ and $J_2 = \int_c^\pi \log|y_s|\, d\theta$, direct computation gives

$$\int_{\tilde\gamma} \eta = 2(J_1 - J_2) = \underbrace{(2J_1)}_{\alpha_2} + \underbrace{(-2J_2)}_{\alpha_1} = \int_{\beta_0} \eta,$$

hence the *exact* identity

$$\int_{C'} \eta = 2\int_{\tilde\gamma} \eta. \tag{3}$$

4.6. **Regulator constants: Bloch, Brunault, Bertin.** The divisor algebra (projective computation on the minimal cubic model $y^2+y = x^3-x^2$, to which the data are transported by the explicit birational change of variables; Abel's principal-divisor test passed) gives, with $A = (0,0)$ the 5-torsion point and $O = [0:1:0]$,

$$\mathrm{div}(x) = [A] + [2A] - [O] - [3A], \qquad \mathrm{div}(y) = 3[A] - 2[O] - [3A],$$

hence the diamond product

$$(x) \diamond (y) = 6(O) + 5(A) - 5(2A), \qquad D_E\big((x)\diamond(y)\big) = 5D_E(P) - 5D_E(2P).$$

The elliptic dilogarithm values (mpmath, 60 digits; lattice basis $(w_1, w_1 - w_2)$ so that $\tau = 0.5 + 0.2299i$; $z(P) = 0.6w_1$, $z(2P) = 0.2w_1$ from PARI `ellpointtoz`):

$$D_E(P) = 0.191193737084331695754954434312173816 1012\ldots$$

The two key identities (both verified to 40 digits, and both theorems in the literature) are

- **Brunault's coefficient** [5]*(3.151): $D_E(P) = \dfrac{2\pi}{5} b_{11} = \dfrac{11}{10\pi} L(E,2)$, i.e. $L(E,2) = \frac{10\pi}{11} D_E(P)$;
- **Bertin's exotic relation** [1]: $D_E(2P) = \frac{3}{2} D_E(P)$ (note $\frac{3}{2}$, not the naive 2).

Closing the loop:

$$5D_E(P) - 5D_E(2P) = 5\Big(1 - \tfrac{3}{2}\Big) D_E(P) = -\tfrac{5}{2} D_E(P) = -\pi b_{11}.$$

By Bloch's theorem [3] in its original normalisation $r(\{f,g\}) = \frac{1}{2\pi}\int_\gamma \eta(f,g)$ (γ generating $H_1(E,\mathbb{Z})^-$) one has $\pi\, r(\{f,g\}) = D_E((f)\diamond(g))$ [18]*Thm. 1, i.e.

$$\int_{\gamma^-} \eta = \pm 2\, D_E\big((x)\diamond(y)\big) = \mp 2\pi b_{11},$$

the sign depending on orientation (Brunault, Remarque 20: D_E depends on the orientation of $E(\mathbb{R})$ and is defined only up to sign). In fact Brunault's (3.210)–(3.211), quoting Bertin, already give $|r_{\gamma^-}\{x,y\}| = \frac{5}{2\pi} D_E(P) = b_{11}$

directly: the regulator side is a proved theorem; the gap closed here is the identification of Boyd's split-integral chain with the closed cycle in $H_1(E,\mathbb{Z})^-$.

4.7. **Synthesis.**

Theorem 4.3 ((C3))**.** $I_{\text{split}} = b_{11}$.

Proof. By (3) and $\text{class}(C') = 2\gamma^-$ (§4.4),

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2}\int_{C'} \eta = \frac{1}{2}\cdot 2 \cdot (\pm 2\pi b_{11}) = \pm 2\pi b_{11}.$$

The sign is pinned by a single numerical evaluation: $\int_{\tilde{\gamma}} \eta = +0.9559686854\ldots = +2\pi b_{11}$ with $b_{11} > 0$. Finally, by the identity $I_{\text{split}} = \frac{1}{2\pi}\int_{\tilde{\gamma}} \eta$ of §4,

$$I_{\text{split}} = b_{11}.$$ □

5. The conductor-17 case: $k = 1$

The proof of (C3) in §4 is a *method*, and its first re-application is Samart's conductor-17 analogue. For $k = 1$ the polynomial

$$S_1 = y^2 + (x^2 + x + 1)y + x^3$$

defines an elliptic curve of conductor 17 (PARI `ellfromeqn` gives $E_1 : y^2 + xy - y = x^3 - x^2$, $\Delta = 17$; `ellglobalred` confirms the conductor), with $E_1(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/4\mathbb{Z}$ generated by $A = (0,0)$, which has order 4 — the exact parallel of the 5-torsion point of the $k = 0$ case. Every step of §4 goes through, and the regulator side is in fact *easier*: the key D_E-value is a published theorem of Lalín–Ramamonjisoa [10].

5.1. **The closed cycle.** The torus-intersection (fold) angle is $c = \arccos(-k/2)\big|_{k=1} = 2\pi/3$. At $\theta = c$, $x = \omega = e^{2\pi i/3}$ satisfies $x^2 + x + 1 = 0$, so S_1 reduces to $y^2 + 1 = 0$: the branch jump values are *exactly* $y = \pm i$ (the $k = 0$ analogue was $y^2 = i$). Writing $P = (\omega, i)$, the signed chain $\tilde{\gamma}$ (big branch on $[-c, c]$ minus the big branch on the two outer arcs) has boundary

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}],$$

and the same small-branch compensating arc β_0 closes it to a cycle $C' = \tilde{\gamma} + \beta_0$ with $\partial C' = 0$, $c(C') = -C'$.

One structural difference from $k = 0$: here $\Delta = +17 > 0$, so $E(\mathbb{R})$ has two connected components, $\overline{w_2} = -w_2$ holds on the nose, and the primitive anti-invariant period is simply

$$w_{\text{anti}} = w_2 = -2.7457391180897536720341879\ldots\, i$$

(for $k = 0$, $\Delta < 0$ forced $w_{\text{anti}} = 2w_2 - w_1$).

5.2. **Homology class: interval-arithmetic proof.** As in §4.4, period$(C')/w_{\mathrm{anti}}$ is a-priori a non-zero integer. The Arb certification (script `code/k1_interval.py`; output `attack11-k1-interval.txt`, 300-bit working precision) is simpler than for $k=0$ because $D(z)=z^4-2z^3+3z^2+2z+1$ has *no zeros on* $|z|=1$ ($\min_{|z|=1}|D|=4>0$, certified): there is no endpoint singularity and no tip-estimate machinery. The period w_{anti} is certified independently of PARI via Newton–Rouché isolation of the roots of $4x^3-3x^2-2x+1=(x-1)(4x^2+x-1)$ and the two-component Carlson R_F formulas. The resulting ratio ball is

$$[-2.00\pm 2.4\times 10^{-14}]+[\pm 2.4\times 10^{-14}]\,i, \qquad |\mathrm{ratio}+2|\le 3.4\times 10^{-14}<\tfrac{1}{2},$$

(the radius is set by the integration tolerance, not by the arithmetic). Choosing γ^- with period$(\gamma^-)=w_{\mathrm{anti}}$, the a-priori integrality upgrades to the exact equality

$$\mathrm{class}(C')=2\gamma^- \qquad (\text{the script uses } -w_{\mathrm{anti}},\ \text{displaying } -2).$$

5.3. **The regulator side.** The integral algebra of §4 is unchanged: the pointwise identity $\log|y_{\mathrm{big}}|=-\log|y_{\mathrm{small}}|$ on $|x|=1$ gives $\int_{\beta_0}\eta=\int_{\tilde\gamma}\eta$, hence $\int_{C'}\eta=2\int_{\tilde\gamma}\eta$. The divisors have the same shape

$$\mathrm{div}(x)=[A]+[2A]-[O]-[3A], \qquad \mathrm{div}(y)=3[A]-2[O]-[3A],$$

now verified *four* ways: local expansions, PARI, code expansion, and an explicit birational map $X=-(x+y)$, $Y=x(x+y)$ from S_1 to E_1 : $Y^2+XY-Y=X^3-X^2$. The diamond product (definition as in [11]) expands exactly to

$$(x)\diamond(y)\equiv 6(O)+4(A)-6(2A) \qquad \text{in } \mathbb{Z}[E]^-.$$

The two D_E-values are settled:

- $D_E(2A)=0$ *rigorously*: $2A$ is 2-torsion, q is a positive real, and the Bloch–Wigner series vanishes termwise at $z_q=-1$;
- $D_E(A)=\dfrac{17}{8\pi}L(E_1,2)$ is a *published theorem*: Lalín–Ramamonjisoa [10]*§5 prove $L(E_{17},2)=\frac{8\pi}{17}D^E(P)$ for the 4-torsion generator P of their model, in the D^E-normalisation (their Def. 5, eq. (10)) that coincides verbatim with our series implementation; we reproduced the value independently to 60 digits.

Hence

$$D_E\big((x)\diamond(y)\big)=4D_E(A)=\frac{17}{2\pi}L(E_1,2)=2\pi b_{17},$$
$$b_{17}=L'(E_1,0)=\frac{17}{4\pi^2}L(E_1,2).$$

5.4. **Synthesis.** We use Bloch's theorem in the normalisation of [10]*Thm. 6: for $\{x,y\} \in K_2(E)\otimes\mathbb{Q}$ and γ^- generating $H_1(E,\mathbb{Z})^-$,

$$\int_{\gamma^-} \eta(x,y) = \pm D^E\big((x)\diamond(y)\big).$$

The temperedness hypothesis holds: the four face polynomials of S_1 are x^3+x^2y, x^2y+y^2, x^3+y^2 and $y(x^2+x+1)$, all cyclotomic since $x^2+x+1 = \Phi_3(x)$. This normalisation has published precedent on this very curve: the conductor-17 identity of [10] combined with Zudilin's $m = 2b_{17}$ closes with the same factor.

Theorem 5.1 (Samart's conductor-17 analogue)**.** $\tilde{n}(1) = b_{17}$.

Proof. By the above,

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2}\int_{C'} \eta = \frac{1}{2}\cdot 2\cdot\big(\pm D_E((x)\diamond(y))\big) = \pm 2\pi b_{17},$$

and the sign is pinned by a single numerical evaluation:

$$\int_{\tilde{\gamma}} \eta = -1.8809066251248561\ldots = -2\pi b_{17}.$$

The structural identity $\tilde{n}(1) = -\frac{1}{2\pi}\int_{\tilde{\gamma}}\eta$ (the $k=1$ instance of the pointwise argument of §4) gives $\tilde{n}(1) = b_{17}$. ☐

Remark 5.2. The $k = 0$ proof (§4) quotes Bloch's theorem through Brunault's regulator r_γ, with the factor $\int\eta = \pm 2D_E(\cdot)$ certified there by Brunault's (3.210)–(3.211); the present proof quotes the Lalín normalisation with factor 1, certified here by the published conductor-17 identities of [10] and [19]. The two formulations differ only in the conventional factor in the definition of r_γ versus D^E; each is used with the published formula that pins its normalisation on the respective curve.

6. Numerical certification

6.1. **Certified computations.** All floating-point computations are performed with mpmath at 60–300 digits and cross-checked with PARI/GP 2.15.5:

- **The constant** b_{11} (`code/b11.py`): from the exact integer coefficients of $f_{11} = \eta(\tau)^2\eta(11\tau)^2$ (truncated convolution of the squared Euler function) via the approximate functional equation for weight 2, root number $+1$:

$$b_{11} = \Lambda(f,2) = \sum_{n\geq 1} a_n\Big[e^{-t_n}\Big(\frac{1}{t_n}+\frac{1}{t_n^2}\Big) + E_1(t_n)\Big], \qquad t_n = \frac{2\pi n}{\sqrt{11}},$$

  with terms decaying like $e^{-1.894n}$; 200 terms suffice for 300 digits.

- **(C3) to** 149 **digits** (`code/attack3.py`):

$$\begin{aligned} I_{\text{split}} = 0.&15214714172591804948622729747863449562814 \\ &3589164226122809889823882023289695302776676\ldots, \end{aligned}$$

$$|I_{\text{split}} - b_{11}| = 4.85 \times 10^{-149}.$$

- **(C1)/(C2) reproduced** to 52 digits (`code/attack1.py`): $\|m - 7b_{11}\| \approx 5.0 \times 10^{-53}$, $\|m - 5b_{11}\| \approx 3.6 \times 10^{-53}$.
- **Period certification** (`code/n1_certify.py`, `code/n1_certify.gp`): (2) computed independently at 60 and 80 digits with identical results; each arc integral A_s is stable to 24 digits across 40/60/80-digit runs; w_{anti} taken from PARI `E.omega` for `11.a3`. The integrality argument has an a-priori gap of 1 against a residual of 2.5×10^{-16}.
- **Exact algebraic checks**: group law and torsion (`code/torsion.py`, `code/endpoint_torsion2.py`) use exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$—no numerical approximation; the model constant $\kappa = 1$ is fixed by the discrete candidates $\kappa^{12} = \Delta_{\text{quartic}}/\Delta_{\min} = 2^a 11^b$ plus 50-digit agreement.
- **PARI/GP cross-checks**: `verify_family.gp`, `verify_ratios.gp` (family conductors, j-invariants, b-values vs. `lfun` to 50+ digits), `winding.gp`, `dilog.gp`, `k53.gp`, `k53b.gp`, `kfamily_torsion.gp`.
- **Conductor-**17 **certification** (scripts `code/k1_*`, listed individually in Appendix A): PARI curve identification and z-values; mpmath 60/80-digit chain integrals (ratio 2.0000000000000002); exact diamond-product expansion with the $k = 0$ case as control; $D_E(A)$, $D_E(2A)$ to 60 digits; and the Arb ironclad proof of the period ratio (§5).

6.2. **Level of rigour: interval certification.** The integer identification in §4.4 has been made fully interval-rigorous (script `code/n1_interval.py`, python-flint/Arb ball arithmetic [8], 300-bit working precision; output in `notes/attack10-interval.txt`). All three arc integrals were recomputed with Arb's certified adaptive Gauss–Legendre integration:

- the endpoint singularity $u \sim \sqrt{\theta}$ at $\theta = 0$ is removed by the substitution $\theta = \pm t^2$; the substituted integrand is analytic on the disc $|t| \le \rho = 0.6$ (the nonzero zeros of $D(t^2)$ satisfy $|\theta_j| \ge 1.2188$, certified by Newton–Rouché isolation of the roots of $z^4 - 4z^3 + 2z^2 + 1$), and the tip $[0, \delta]$, $\delta = \rho/5$, is summed by the Cauchy estimate $\int_0^\delta f = a_0\delta + R$, $|R| \le H\delta^3/3$, where H is derived from $M = \max_{|t|=\rho} |f| = 1.3075\ldots$, itself certified by covering the circle with 4096 balls (per-tip Cauchy remainder bound $H\delta^3/3 = 2.18 \times 10^{-3}$);
- $D(\theta)$ crosses the negative real axis once on (c, π) and once on $(-\pi, -c)$, where the principal square root is not analytic: the arcs are subdivided adaptively, on each piece either $\sqrt{D}$ or $i\sqrt{-D}$

is certified analytic by ball evaluation, and the overall sign is propagated across the nodes by a certified matching test;
- w_{anti} is certified independently of PARI: the roots of $4x^3 - 4x^2 + 1$ are isolated by Newton iteration plus a Rouché certificate, and Carlson's R_F (DLMF §19) gives $w_{\mathrm{real}} = 6.34604652139776710844397\ldots$ and the purely imaginary period $2.91763323387699045866178\ldots i$ — which is $-w_{\mathrm{anti}}$ in the convention of §4.4 — with certified radii 6.7×10^{-48} and 4.9×10^{-49}, agreeing with PARI to 45 digits.

Against w_{anti} the resulting ratio ball is $[2.00 \pm 3.13 \times 10^{-3}] + [\pm 2.99 \times 10^{-3}]\, i$: it contains 2 and $|\text{ratio} - 2| \leq 4.33 \times 10^{-3} < 1/2$. The a-priori integrality of §4.4 therefore upgrades to the exact equality $\mathrm{period}(C') = 2\, w_{\mathrm{anti}}$, i.e. $\mathrm{class}(C') = 2\gamma^-$. (The script itself works with the opposite sign of the period throughout, so the raw output in `notes/attack10-interval.txt` displays the ratio ball centred at -2; the two formulations are equivalent.)

7. Conjectures

The family identities established numerically here are stated as conjectures pending analytic proof:

Conjecture 7.1. *For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ with $k \notin (-4, 2)$ (integer k), there is a small rational number r_k such that*

$$m(S_k) = r_k\, |L'(E_k, 0)|.$$

Confirmed numerically: $r_2 = 2$, $r_3 = 1$ (60 digits), and $r_{-4} = \frac{7}{2}$ (conductor 37), $r_{-5} = \frac{1}{4}$ (conductor 359), $r_{-6} = \frac{1}{8}$ (conductor 997) (25 digits each).

The last three deserve emphasis as *predicted then confirmed*: the structural theorem (Theorem 3.4) predicted, from the torus-intersection trichotomy, that $k = -4, -5, -6$ must satisfy Boyd-type identities with specific small rationals, and the subsequent computation hit the predicted values. Note that S_{-4} and S_2 share conductor 37, yielding Rodríguez Villegas–type rational relations [14, 10] between two different polynomial models of the same curve.

Remark 7.2 (Samart's conductor-17 analogue: now a theorem)*.* The conjecture $\tilde{n}(1) = b_{17}$ for $k = 1$ (conductor 17), suggested by Samart and confirmed numerically to 60 digits, is proved as Theorem 5.1 in §5.

Remark 7.3*.* By Theorem 3.6, no such conjecture can be formulated for $k = -1$ (conductor 53): the anti-invariant cycle on the torus does not exist and x, y are not modular units.

Appendix A. Reproduction code

All scripts live in `code/`; raw outputs in `notes/attack*.txt`. Dependencies: Python 3.12 with mpmath and sympy; PARI/GP 2.15.5; python-flint 0.9.0 (Arb) for the interval certification.

File	Purpose
`b11.py`	$b_{11} = L'(E_{11}, 0)$, 300 digits (functional equation)
`attack1.py`	reproduce (C1), (C2) to 52 digits
`attack2.py`	$m(S_0)$, PSLQ negative results
`attack3.py`	(C3) to 149 digits
`torsion.py`	exact group law: $5A = O$, modular units
`endpoint_torsion2.py`	exact check: endpoint P non-torsion
`boundary_torsion.py`	boundary-divisor torsion analysis
`closedness_check.py`	$\tilde{n} = -I_{\text{split}}$, regulator $= 2\pi I_{\text{split}}$
`ntilde_family.py`	$\tilde{n}(k)$ family table, 50–80 digits
`b_family.py`	b_N for the family via point counting
`winding.py` / `winding.gp`	period pairings, winding-number heuristics
`dilog.py` / `dilog.gp`	elliptic dilogarithm $D_E(P)$, Brunault constants
`k53_attack.py`	conductor-53 obstruction analysis
`k53.gp` / `k53b.gp`	PARI checks for `53.a1`
`kneg_m.py`	$m(S_k)$ for $k = -4, -5, -6$ (predicted identities)
`n1_certify.py` / `n1_certify.gp`	certification of class$(C') = 2\gamma^-$
`n1_interval.py`	Arb ironclad proof of the period ratio
`branch_certify.py`	branch assignments + modulus ordering (certified)
`k1_pari/points/zvals.gp`	conductor-17 identification, torsion, z-values
`k1_certify.py`	mpmath certification of the $k = 1$ chain integrals
`k1_interval.py`	Arb ironclad proof of the $k = 1$ period ratio
`k1_diamond.py`	exact diamond product for $k = 1$ ($k = 0$ control)
`k1_dilog.py`	$D_E(A)$, $D_E(2A)$ for conductor 17
`verify_family.gp`	family conductors and j-invariants
`verify_ratios.gp`	PARI `lfun` cross-check of ratios
`kfamily_torsion.gp`	torsion, ord$(0, 0)$ for the family

Reproduction:

```
cd code && python b11.py && python attack1.py && python attack2.py \
  && python attack3.py && python torsion.py && python endpoint_torsion2.py \
  && python boundary_torsion.py && python closedness_check.py \
  && python ntilde_family.py && python b_family.py && python winding.py \
  && python dilog.py && python k53_attack.py && python kneg_m.py \
  && python n1_certify.py
.venv/Scripts/python n1_interval.py    # Arb certification, k=0
.venv/Scripts/python branch_certify.py  # branch assignments + mod ordering
.venv/Scripts/python k1_interval.py     # Arb certification, conductor 17
gp -q verify_family.gp && gp -q verify_ratios.gp
gp -q winding.gp && gp -q dilog.gp
gp -q k53.gp && gp -q k53b.gp && gp -q kfamily_torsion.gp
gp -q k1_pari.gp && gp -q k1_points.gp && gp -q k1_zvals.gp
```

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